Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*			**Indicative content:** 1. Measure the mass of the block with a balance. 2. Put sufficient water e.g. 20 cm^3 into a measuring cylinder 3. Record the volume. 4. Place the metal in the water and ensure it is fully submerged 5. Record the new volume. 6. Calculate the volume of the metal by subtracting volume of water from volume of water + metal. 7. Calculate the density by dividing mass by volume. **5 – 6 marks** Fully describes a workable method, including statement 7 *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3 – 4 marks** Identifies relevant apparatus and describes a method to allow density to be calculated. Uses measuring cylinder method for volume but misses some detail or misses detail on how to calculate results. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Mentions some relevant apparatus or measurements or refers to volume of a regular shape *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*	6			6		6

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Iron (1) It is the clos<u>est</u> value [or equiv, e.g. <u>most</u> similar] given (1) N.B. 2nd mark only available if the first mark is awarded.			2	2		2
		(ii)		Any one for (1) of • Value is in between 2 metals • Value is not very close to one metal • Could be another metal not listed • None of the metals have the exact density • Tin and/or copper have a similar density [accept 'iron has a similar density' if tin or copper given in *(b)*(i)]			1	1		1
		(iii)		Use higher resolution measuring cylinder / higher resolution balance Or measure {volume / mass} to {a higher resolution / more significant figures} [Accept: measure to a higher resolution]			1	1		1
	(c)	(i)		Answer = 400 [cm^3]		1		1	1	1
		(ii)		Substitution: 19.32 × 400 **ecf** (1) Mass = 7 728 [g] (1)	1	1		2	2	2
				Question 3 total	**7**	**2**	**4**	**13**	**3**	**13**